

A Study of Difference between the Scores Awarded to the Gymnasts during Domestic Competitions/ Selection Trials and World Gymnastics Championships.

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ABSTRACT

The present study was conducted to find out the differences between the Scores awarded to the 5 Indian Women Gymnasts during the Selection/ Domestic competitions for participating in the World Gymnastics Championships in 2019 and 2017 and the Scores they obtained during the World Gymnastics Championships in 2019 and 2017. The results of the investigation revealed that there was too much difference between the Scores awarded to the 5 Indian Women Gymnasts during the Selection/ Domestic competitions for participating in the World Gymnastics Championships in 2019 and 2017 when compared with the Scores they obtained during the World Gymnastics Championships in 2019 and 2017. It was found that the Indian Gymnastics Judges award too high Scores during domestic competitions when compared with the Scores the same gymnasts received during International Competitions.

INTRODUCTION

In Artistic sports like Women Artistic Gymnastics (WAG), the performance in competition is evaluated by judges that apply a tool (Code of Points) and give a Score that determines the value of the Routine and the position of the Gymnast in the final ranking. Since the performance does not come out from an objective measure, but from a complex judging process, quite often Gymnastics is considered to be a subjective sport (Gateva, 2014).

Judging in Artistic Gymnastics crucially influences sport results. The differences between competitors are often small, especially if the homogenous group such as the world class gymnasts competes at higher level

competitions (European and World championships, Olympic Games) (Gymnastics Results Com, 2012). Here even a small systematic bias of judges may influence final ranks of competitors. Therefore, continuous monitoring of the quality of judging (incorporating reliability and validity) is a necessity.

Recent research has paid attention mainly to the experience and the capacity of the judges to use cognitive and perceptual strategies to interpret and register Gymnast's performance in competition (Dallas and Kirialanis, 2010; Heinen et al., 2012; Plessner and Schallies, 2005; St. Marie et al., 2001). Furthermore, research has also emphasized the judge's need for

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developing a set of skills that contributes to an effective assessment process (Fernandez-Villarino et al., 2013) and to the overall error detection efficiency (Flessas et al., 2015).

International Gymnastics Federation (FIG) has prepared Code of Points for Women Artistic gymnastics to evaluate each performance at International Competitions.

As per the present Code of Points for WAG (FIG, 2017-2020), there are two Panels of Judges which evaluate each performance. One Panel, which is known as "D" Jury, evaluates the "D" Score. "D" Jury consists of two Judges, i.e. "D1" and "D2". "D" Score is the Content Value of a Routine, which comprises of the three factors: i.e., (a) Difficulty Value, (b) Compositional requirements (which vary by apparatus) and (c) Connection Value.

The Second Panel of Judges is known as "E" Jury, which comprises of 5 Judges, known as "E 1", "E 2", "E 3", "E 4" and "E 5". "E" Panel evaluates routines for Execution, Technique and, for the women, Artistry. Each "E" judge starts with 10.00 points and then makes deductions for any committed faults in those areas. This is the panel size for FIG-conducted international events; it may vary at domestic events. In addition to the 5 "E" Judges, there are also 2 Reference Judges.

In Artistic Gymnastics, in International competitions, the Execution Score is first calculated with

the trimmed mean of the five Execution Panel Judges. This Score is then compared to the reference Score consisting of the mean Score of the two Reference Judges. If the gap between the Execution Panel Score and the Reference Score exceeds a predefined tolerance, the Final Score of the athlete is the mean of the Execution Panel and Reference Scores.

In spite of such hard and strict rules of the FIG, there comes a lot of differences in the Scores awarded by the Judges. According to Flessas et al (2014). Judging a gymnastics routine is a noisy process, and the performance of judges varies widely. Judging is much more about skill and training than bias: it is challenging to evaluate every single aspect of the complex movements that are part of a Gymnastics routine". Sometimes the Judges commit mistakes and award higher or lower Scores. Fink (2020), has reported that the Gymnastics Judges commit the mistake and award higher or lower scores because of the 4 "Cs", i.e., "": Lack of Conscience; Lack of Control; Lack of Confidence; Lack of Competence. A fifth "C" could be Lack of (an adequate) Code of Points.

There has always been a doubt on the quality of Gymnastics Judging during the National Competitions and Selection trials. It has been observed many a times that the Indian Gymnasts obtain less Scores in International Competitions when compared with the

Scores they obtain in Domestic Competition or Selection Trials. This may be because of the poor judgment at Domestic level competitions. The Indian Judges are quite liberal in awarding Scores during Domestic competitions / selection trials. The present investigation has been conducted with a purpose to find out the difference in the Scores the Indian gymnasts obtain during the International Competitions and domestic competitions/selection trials.

METHODOLOGY:

The present investigation was conducted on the Scores awarded to the three India Women Gymnasts who participated in the World Gymnastics Championships, held in 2017 in Canada and 2019 World Championships held in Germany. Their Scores were obtained from the official results of the World Gymnastics Championships. The Scores awarded to them during the Selection

Trials conducted about one month prior to the World Championships were collected from the official results of the Selection Trials. The Scores awarded to the Russian and US Gymnasts during their Domestic competitions and 2016 Rio Olympics were also collected from the official results of these competitions to find out the difference between the judgement of the Indian Judges and Russian and US Women Gymnastics Judges.

RESULTS & DISCUSSION:

While comparing the Scores awarded to the three gymnasts during the World Gymnastics Championships and Selection Trials, it has been found that the Indian Judges awarded extremely higher Scores in Domestic Competition/ Selection Trials. The scores achieved by the Gymnasts during the Selection Trials and World Championships are presented in the following Tables.

Table-1
Comparative Study of the Scores awarded to the Three Indian Gymnasts during the Selection Trials held on 16-09-2019 in Delhi and during the 49th World Gymnastics Championships held in Oct. 2019 in Germany

SNO	EVENT/ NAME	1 st Ranking Gymnast			2 nd Ranking Gymnast			3 rd Ranking Gymnast		
		MARKS DURING SELECTION TRIALS FOR WORLD CHAMPIONSHIPS HELD ON 16, SEPT. 2019	MARKS IN WORLD CHAMPIONS SHIPS, 4-13 OCTOBER, 2019 IN STUTTGART	DIFFERENCE	MARKS DURING SELECTION TRIALS FOR WORLD CHAMPIONSHIPS HELD ON 16, SEPT. 2019	MARKS IN WORLD CHAMPION SHIPS, 4-13 OCTOBER, 2019 IN STUTTGART	DIFFERENCE	MARKS DURING SELECTION TRIALS FOR WORLD CHAMPIONSHIPS HELD ON 16, SEPT. 2019	MARKS IN WORLD CHAMPIONSHIPS, 4-13 OCTOBER, 2019 IN STUTTGART	DIFFERENCE
1	VAULT	14.20	13.80	0.40	13.35	13.00	0.35			
2	UNEVEN BARS	12.20	10.56	1.64	11.00	9.916	1.084	11.40	8.925	2.475
3	BEAM	12.20	9.993	2.207	12.450	10.866	1.584	11.70	10.200	1.50
4	FLOOR	12.55	11.133	1.417	12.30	11.466	0.834			
5	INDIVIDUAL ALL AROUND SCORES	50.75	45.832	4.918	49.10	45.248	3.852	23.10	19.125	3.975

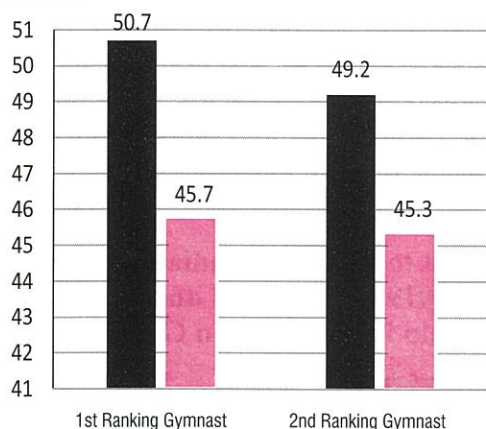
Table -2

Comparative Study of the Scores awarded to the Two Indian Gymnasts during the Selection Trials held on 26th& 27th August in Delhi and during the 48th World Gymnastics Championships held in Oct. 2017 in Canada (Only 2 Gymnasts participated)

SNO	EVENT/ NAME	1st Ranking Gymnast			2nd Ranking Gymnast		
		MARKS DURING SELECTION TRIALS, HELD ON 26 & 27 AUG, 2017	MARKS IN 2017 WORLD CHAMPIONSHIPS, 3 OCTOBER, 2017 IN MONTREAL	DIFFERENCE	MARKS DURING SELECTION TRIALS, HELD ON 26 & 27 AUG, 2017	MARKS IN 2017 WORLD CHAMPIONSHIPS, 3 OCTOBER, 2017 IN MONTREAL	DIFFERENCE
1	VAULT	13.700	13.40	0.40	13.700	13.433	0.267
2	UNEVEN BARS	11.55	10.333	1.271	11.15	9.733	1.417
3	BEAM	12.95	10.033	2.917	10.55	9.033	1.517
4	FLOOR	12.400	9.10	3.30	12.45	10.666	1.784
5	INDIVIDUAL ALL AROUND SCORES	50.60	42.866	7.734	47.85	42.865	4.985

Graphical Presentation of All Around Scores Obtained by Indian Gymnasts During Domestic Competition In 2019 and in 2019 World Championships

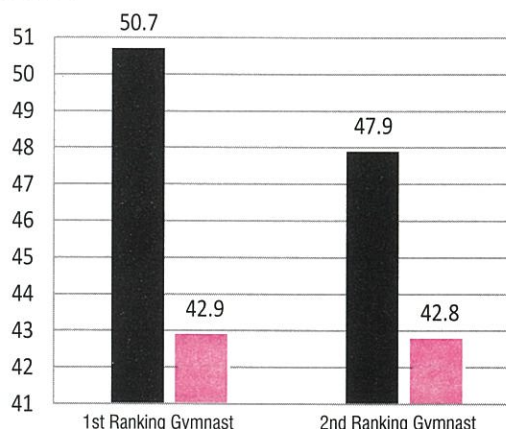
SCORES



■ DC (Domestic Comp.) ■ WC (World Champ.)

Graphical Presentation of All Around Scores Obtained by Indian Gymnasts During Domestic Competition In 2017 and in 2017 World Championships

SCORES



■ DC (Domestic Comp.) ■ WC (World Champ.)

Table-1 depicts the Scores obtained by the three India women Gymnasts during the Selection Trials held in September, 2019 for the selection of the Indian Gymnasts for the 49th World Gymnastics Championship and Scores obtained during the 49th World

Gymnastics Championship held in October, 2019 in Stuttgart (Germany). It is evident from the Table that 1st Ranking Gymnast obtained All Around Scores of 50.75 points during the Selection trials where she obtained 45.832 in the 49th World Gymnastics

Championship. She obtained 4.918 points less in the 49th World Gymnastics Championship when compared with the Scores she obtained during the Selection Trials. The above Table-1 also reveals that she got higher Scores in all the Events during the Selection Trials when compared with the Scores she achieved in the 49th World Gymnastics Championship.

The Table-1 also depicts the similar results in case of the second Gymnast. 2nd Ranking Gymnast obtained All Around Scores of 49.10 points during the Selection trials but she obtained 45.248 in the 49th World Gymnastics Championship. She obtained 3.852 points less in the 49th World Gymnastics Championship when compared with the Scores she obtained during the Selection Trials. The above Table-1 also reveals that she got higher Scores in all the Events during the Selection Trials when compared with the Scores she achieved in the 49th World Gymnastics Championship.

Looking at the Scores of the third Gymnast, 3rd Ranking Gymnast, it is evident that she also obtained 3.975 points lesser during the 49th World

Gymnastics Championship when compared with the Scores she obtained during the Selection Trials.

The results in Table -2 show the similar pattern of difference of Scores obtained by the 2 Gymnasts who participated in the 48th World Gymnastics Championship held in Montreal in 2017. 1st Ranking Gymnast obtained 7.734 points lesser and 1st Ranking Gymnast received 4.985 lesser Scores during the 48th World Gymnastics Championship when compared with Scores they received during the Selection Trials. Table -2 also reveals that both the gymnasts obtained lesser Scores in All the Events during 48th World Gymnastics Championship when compared with Scores they received during the Selection Trials. Such a difference of Scores is extremely great.

The researchers also tried to find out that how much differences existed in the Scores of the Russian and American Women Gymnasts during their participation in the 2019 World Gymnastics Championship and Domestic Selection Competitions. The

Table- 3

Comparative study of the Scores awarded to Two RUSSIAN Gymnasts during the Selection Trials held on 21-27/08/19 and during the 49th World Gymnastics Championships held in Oct. 2019 in Germany

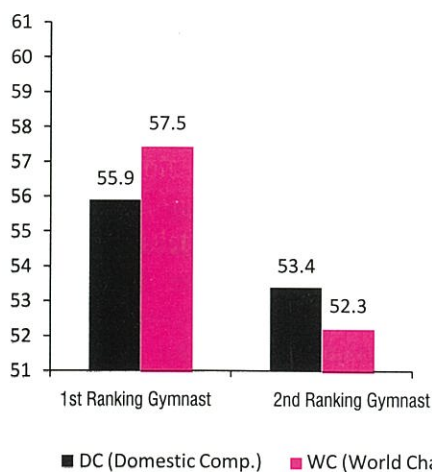
SNO	EVENT/ NAME	1st Ranking Gymnast from Russia			2nd Ranking Gymnast from Russia		
		MARKS DURING SELECTION TRIALS, 21-27/08/19	MARKS IN WORLD CHAMPIONSHIPS , 4-13 OCTOBER, 2019 IN STUTTGART	DIFFERENCE	MARKS DURING SELECTION TRIALS	MARKS IN WORLD CHAMPIONSHIPS , 4-13 OCTOBER, 2019 IN STUTTGART	DIFFERENCE
1	VAULT	14.366	14.938	-.572	14.700	14.737	-0.037
2	UNEVEN BARS	14.533	14.70	-0.167	11.300	13.600	-2.30
3	BEAM	13.433	13.66	-0.227	13.366	12.90	0.466
4	FLOOR	13.600	14.10	-0.5	14.033	13.933	0.10
5	INDIVIDUAL ALL AROUND SCORES	55.932	57.478	-1.546	53.399	55.260	-1.861

Table- 4
Comparative Study of the Scores awarded to Two Gymnasts from USA during the Selection Trials held on 19-20/07/19 and during the 49th World Gymnastics Championships held in Oct. 2019 in Germany

SN	EVENT/ NAME	1st Ranking Gymnast, USA			2nd Ranking Gymnast, USA		
		MARKS DURING U.S CLASSIC GYMNASTICS TOURNAMENT HELD 19-20, JULY, 2019	MARKS IN WORLD CHAMPIONSHIPS , 4-13 OCTOBER, 2019 IN STUTTGART	DIFFERENCE	MARKS DURING U.S CLASSIC GYMNASTICS TOURNAMENT HELD 19-20, JULY, 2019	MARKS IN WORLD CHAMPIONSHIPS , 4-13 OCTOBER, 2019 IN STUTTGART	DIFFERENCE
1	VAULT	15.650	15.066	0.584	14.300	14.566	-0.266
2	UNEVEN BARS	14.450	14.733	-0.283	14.550	14.841	-0.291
3	BEAM	14.900	14.800	0.10	14.100	12.933	1.167
4	FLOOR	15.000	14.833	0.167	14.240	13.766	0.474
5	INDIVIDUAL ALL AROUND SCORE	60.000	59.462	0.538	57.190	55.966	1.224

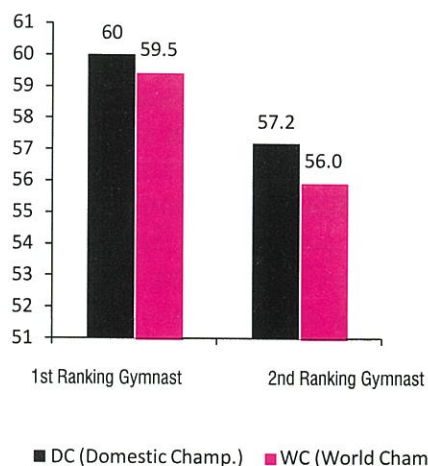
Graphical Presentation of All Around scores of the Russian Women Gymnasts During Domestic competition in 2019 and in World Championship 2019

Scores



Graphical Presentation of All Around scores of the USA Women Gymnasts During Domestic competition in 2019 and in World championship 2019

Scores



It is observed from Table -3 that the two Russian Gymnasts , who participated in all Events in the World Championships obtained slightly less marks in their Domestic / Selection Competition when compared with the Scores they achieved during the 49th World Gymnastics Championships held in Oct. 2019 in Germany. These findings are just opposite to the findings

of the Scores of the Indian Gymnasts. Table-3 reveals that 1st Ranking Gymnast and 2nd Ranking Gymnast, the two Russian Gymnasts obtained 1.546 and 1.861 less Scores respectively in their Domestic/ Selection Competition when compared with the Scores they obtained during the 49th World Gymnastics Championships.

Looking at the Table-4, it is

observed that there was a negligible difference between the Scores obtained by the two US Gymnast during their participation in the Domestic selection Competition and World Gymnastics Championship. 2nd Ranking Gymnast of USA obtained 57.190 points during Selection Competition and 55.966 points during the World Gymnastics Championship. The difference is negligible. 1st Ranking Gymnast of USA, the Olympic and World Gold Medallist, obtained 60.00. points during Selection Competition and 59.462 during the World Gymnastics Championship. The difference of just 0.538 between the two Scores is negligible.

Abdul Razak Abdul Kader (2017) conducted study on the Domestic v/s International Scoring in Gymnastics. He has reported the Scores awarded to the women Gymnasts, from USA, and Russia during National Domestic competitions in 2016 and Rio Olympic Games 2016.

He reported that 1st Ranking Gymnast scored 0.005 points on average higher at home. Though this number is so small. There really was no difference between her home Scores and Olympic Scores. She obtained 62.198 at home and 62.203 in Rio Olympic Games. Gabby Douglas scored 0.085 better on an average at the Olympics. He further reported that Laurie Hernandez scored 0.157 better on an average at home, Madison Kocian scored 0.052 better on average than the Olympic Games on Un-even bars. Aly Raisman scored 0.061 points better in All Around Scores

in Olympics when compared with the Domestic judgment. He concluded that as a whole the US Team scored 0.136 points higher per Routine in Rio Olympics than they did at home..

While describing the Scores of the Russian Gymnasts in Rio Olympics and Domestic Competition, Abdul Razak Abdul Kader (2017) reported that Angelina Melnikova scored 0.089 better in AA Scores in Domestic competition. Alya Mustafina scored 0.292 better in AA Scores in Olympic Games when compared with the home judgement. Maria Paseka scored 0.231 better on Vault at home. Daria Spiridova scored 0.237 better at the Olympic Games when compared with the scores obtained at home. He further concluded that as a whole there was not difference between the Olympic Scores and Domestic Scores in case of Russian Women team. The Russian Team scored only 0.006 points more at home which is literally nothing.

CONCLUSIONS:

From the above results and discussion it may be concluded that the Indian Women Gymnastics Judges award higher Scores to the Gymnasts in Domestic/ Selection competitions when compared with the Scores that the Indian Gymnasts obtain during the International Competitions. There may be another reason that the Indian Women Gymnasts don't give the same performance in the International Competitions which they give during the Selection Trials, which is a matter of further investigation. But such a trend is

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